

PSU 2026 mSCP Workshop Command List and Exercise Solutions

This document shows all of the code that was used during the workshop. Any items with a backslash (“\”) then a new line is meant to be entered on multiple lines, so when typing you can press **Return** after the backslash. If you copy/paste directly, the backslashes and new lines will be respected.

Start Apple Container Subsystem

```
container system start
```

Get the Code - Containers

```
mkdir -p ~/Desktop/mscp/custom
```

Apple Container:

```
container run -it \  
  --volume ~/Desktop/mscp:/mscp/build \  
  --volume ~/Desktop/mscp/custom:/mscp/custom \  
  ghcr.io/usnistgov/mscp_2.0:latest
```

Docker:

```
docker run -it \  
  --volume /Users/<username>/Desktop/mscp:/mscp/build \  
  --volume /Users/<username>/Desktop/mscp/custom:/mscp/custom \  
  ghcr.io/usnistgov/mscp_2.0:latest
```

Generate Guidance (Run in Container)

Default:

```
./mscp.py guidance baselines/macos/cis_lvl2_macos_26.0.yaml
```

Flags:

```
./mscp.py guidance baselines/macos/cis_lvl2_macos_26.0.yaml -s -p -x
```

All:

```
./mscp.py guidance baselines/macos/cis_lvl2_macos_26.0.yaml -A
```

Localization:

```
./mscp.py guidance -L fr baselines/macos/cis_lvl1_macos_26.0.yaml -A
```

Generate Baseline (Run in Container)

List all available baselines:

```
./mscp.py baseline -l
```

Generate cis_lvl2 baseline and view it:

```
./mscp.py baseline -k cis_lvl2  
cat custom/baselines/cis_lvl2_macos_26.0.yaml
```

Generate baseline for macos 15 and view it:

```
./mscp.py --os_name macos --os_version 15.0 baseline -k cis_lvl2
```

```
cat custom/baselines/  
cis_lvl2_macos_15.0_yaml
```

Tailor a cis_lvl2 baseline:

```
./mscp.py baseline -k cis_lvl2 -t
```

Mapping and SCAP

Create a custom mapping from a file:

```
./mscp mapping -c /path/to/mapping.csv
```

Create SCAP/OVAL files:

```
./mscp.py scap -h
```

```
./mscp.py scap
```

Running the Script Manually with Interactive Flags

```
/path/to/compliance.sh --check
```

800-171r2	800-53-r5
3.1.1	AC-2, AC-3, AC-17
3.1.2	AC-2, AC-3, AC-17
3.1.3	AC-4
3.1.4	AC-5

DO NOT RUN THE --FIX or --CFC FLAGS ON YOUR PRODUCTION MACHINE!!!

Exercise Answers

Exercise 1: Generate Guidance

```
./mscp.py guidance baselines/macros/cmmc_lvl2_macos_26.0.yaml -s -p -m
```

Bonus:

```
./mscp.py guidance -l <PATH/T0/LOGO.png> baselines/macros/cmmc_lvl2_macos_26.0.yaml -s -p -m
```

Exercise 2: Tailor baselines

```
./mscp.py --os_name ios baseline -k indigo_base
```

```
./mscp.py baseline -k cmmc_lvl2 -t
```

Exercise 3: Change Rules and Generate Guidance

In container, run:

```
cp rules/system_settings/system_settings_location_services_disable.yaml  
rules/audit/audit_retention_configure.yaml custom/rules/
```

Open the ~/Desktop/mscp folder in Code (or use nano), navigate to the custom rules.

system_settings_location_services_disable.yaml:

```
id: system_settings_location_services_disable  
title: Disable Location Services  
discussion: |  
  We allow users to choose their preference.  
tags:  
  - manual
```

audit_retention_configure.yaml:

```
---  
odv:  
  hint:  
    datatype: string  
    description: Time or disk size, e.g. 60d OR 5G, see the man page for audit_control  
    validation:  
      regex: ^\d+[shdyBKMG](\s(AND|OR)\s\d+[shdyBKMG])?$  
    custom: 15d
```

```
./mscp.py guidance custom/baselines/PSU_Test_cmmc_macos_26.0.yaml -s
```

Bonus:

```
cp src/mscp/data/sections/systemsettings.yaml custom/sections/
```

```
---
name: "System Settings"
description: |
  This section contains the configuration and enforcement of the settings within the macOS System Settings application.
  System Settings used to be known as System Preferences.

  NOTE: The check/fix commands outlined in this section MUST be run by a user with elevated privileges.
```

Open *custom/sections/systemsettings.yaml* in Code, edit, and then rebuild with the same command.